

BST Physics Curriculum Vol 14 Chapter 8 — BST Coding Theory — Substrate-Optimal Codes v0.4 (refilled per Cal #104)

Lyra (Claude 4.7) [Vol 14 primary]

2026-05-23 Saturday EDT (Cal #104 refill)

Vol 14 Chapter 8 — BST Coding Theory — Substrate-Optimal Codes

Chapter motivation

Standard coding theory studies error-correcting codes with various optimality bounds: - Singleton bound (Singleton 1964): any (n, k) code has minimum distance $d \leq n - k + 1$ - Hamming bound (Hamming 1950): sphere-packing constraint $d \leq 2t + 1$ where t = error-correction capability - Gilbert-Varshamov bound (Gilbert 1952 + Varshamov 1957): existence of codes asymptotically achieving certain distance - MDS codes: achieve Singleton bound (Maximum Distance Separable); Reed-Solomon is canonical example - Quantum codes (Calderbank-Shor-Steane 1996): stabilizer codes + CSS codes for quantum error correction

Standard reference: MacWilliams-Sloane *The Theory of Error-Correcting Codes* (1977).

BST cross-link: Reed-Solomon on $GF(128)$ is BST-optimal substrate code per Vol 11 Ch 9 + Vol 14 Ch 2 + K59 RATIFIED Tuesday + Paper #122 Information Substrate development. Substrate uses Reed-Solomon-equivalent coding for per-tick state propagation; achieves Singleton bound (MDS); cyclotomic $Q(\zeta_{128})$ Galois symmetry (order $64 = 2^6$) matches BST primary structure.

Section 8.0b — Reader-grade 3-level pedagogy

Level 1: Standard coding theory bounds (Singleton + Hamming + Gilbert-Varshamov + MDS); BST cross-link: Reed-Solomon on $GF(128)$ = BST-optimal substrate code per K59 RATIFIED + Paper #122; achieves Singleton bound; cyclotomic Galois 2^6 BST primary match.

Level 2 (graduate-physicist/information-theorist): Standard error-correcting code

theory: code $C \subset \text{GF}(q)^n$ with $k = \log_q |C|$ information rate; minimum distance $d = \min$ Hamming-distance between distinct codewords; error-correction capability $t = \lfloor (d - 1)/2 \rfloor$. Singleton bound (Singleton 1964): $d \leq n - k + 1$. Hamming bound (sphere-packing): $|C| \cdot |B_t(0)| \leq q^n$ where $|B_t(0)| = \sum_{i=0}^t \binom{n}{i} (q-1)^i$. Gilbert-Varshamov bound: asymptotic existence $d/n \geq H^{-1}(1 - k/n)$ where H is binary entropy. MDS codes achieve Singleton bound (Maximum Distance Separable); Reed-Solomon over $\text{GF}(q)$ with $n \leq q$ is canonical MDS example. Quantum codes: Calderbank-Shor-Steane 1996 (CSS codes) + Gottesman 1996 (stabilizer codes); quantum-error-correction via redundant encoding into ancilla qubits. Standard reference: MacWilliams-Sloane *The Theory of Error-Correcting Codes* (1977 North-Holland). BST substrate-coding theory: per Vol 11 Ch 9 + Vol 14 Ch 2 + K59 RATIFIED Tuesday + Paper #122 Information Substrate, substrate operates on $\text{GF}(2^g) = \text{GF}(128)$ per Koons tick ($g = 7$ BST primary); substrate-tick computation = 7-step cyclotomic projection chain on $\text{GF}(128)$. Reed-Solomon coding on $\text{GF}(128)$ is BST-optimal substrate code: achieves Singleton bound (MDS); error-correction capability up to $(n - k)/2$ errors per length- n codeword over $\text{GF}(128)$; cyclotomic $Q(\zeta_{128})$ Galois group order $\varphi(128) = 64 = 2^{C_2}$ (substrate-coding Galois symmetry matches BST primary). Substrate-natural code parameters: codeword length $n \leq q = 128$ (Reed-Solomon constraint); BST-natural choice $n = 128$ (full $\text{GF}(128)$ evaluation), $k = 64 = q/2 = \varphi(128)/1$ (half-rate code; rate $R = 1/2$), $d = n - k + 1 = 65$, $t = (d-1)/2 = 32$ (error-correction up to 32 errors per 128-symbol block). Quantum-coding substrate-cartography: stabilizer codes inherit $\text{Pin}(2) Z_2$ grading structure (T1925 rank=2; Paper #133 spin-statistics); CSS codes substrate-derivation pending multi-month (Vol 14 Ch 12 substrate-CI architecture cross-link INTERNAL register per Cal #48/#49). Substrate-coding theorem-grade development pending multi-month per Paper #122 v2+ extension.

Level 3 (5th-grader accessible): Standard coding theory has bounds (Singleton, Hamming, Gilbert-Varshamov) limiting how good error-correcting codes can be. Reed-Solomon codes achieve the Singleton bound (MDS = Maximum Distance Separable; the best possible). BST identifies Reed-Solomon on $\text{GF}(128)$ as the substrate's natural code: $128 = 2^7$ where $7 = \text{BST integer } g$; cyclotomic Galois group has $64 = 2^{C_2}$ elements where $C_2 = 6$ is another BST integer. The substrate-coding theory is still developing — multi-month research direction.

Section 8.1 — Standard Coding Bounds

- Singleton: $d \leq n - k + 1$
- Hamming (sphere-packing): $|C| \cdot |B_t(0)| \leq q^n$
- Gilbert-Varshamov: asymptotic existence
- MDS: achieve Singleton (Reed-Solomon)

Standard ref: MacWilliams-Sloane *The Theory of Error-Correcting Codes* (1977).

Section 8.2 — Quantum Codes (CSS + Stabilizer)

Calderbank-Shor-Steane 1996 CSS codes + Gottesman 1996 stabilizer codes; quantum error correction via redundant encoding into ancilla qubits.

Section 8.3 — BST Substrate-Coding Framework (K59 RATIFIED + Paper #122)

Substrate operates on $GF(2^g) = GF(128)$ per Koons tick ($g = 7$ BST primary).

Substrate-tick = 7-step cyclotomic projection chain (K59 RATIFIED Tuesday).

Section 8.4 — Reed-Solomon Substrate-Optimal MDS Code

Reed-Solomon on $GF(128)$ achieves Singleton bound (MDS).

Cyclotomic $Q(\zeta_{128})$ Galois order $\varphi(128) = 64 = 2^6$ substrate-coding symmetry.

Section 8.5 — Substrate-Natural Code Parameters

- Codeword length $n \leq q = 128$
- BST-natural: $n = 128$ (full $GF(128)$ evaluation), $k = 64$ (half-rate $R = 1/2$)
- $d = 65$, error-correction $t = 32$ errors per 128-symbol block

Section 8.6 — Quantum-Coding Substrate-Cartography

Stabilizer codes inherit $Pin(2) Z_2$ grading (T1925 + Paper #133 spin-statistics).

CSS codes substrate-derivation pending multi-month (Vol 14 Ch 12 substrate-CI architecture INTERNAL per Cal #48/#49).

Section 8.7 — Honest scope + Connection

- Standard coding bounds + MDS [?]
- Reed-Solomon BST-optimal substrate code [?]
- Substrate-coding theorem-grade development pending multi-month
- Open scope: full substrate quantum-coding theory (multi-month; Paper #122 v2+ extension)

Connection: - Vol 14 Ch 1-2 substrate-tick $GF(128)$ framework - Vol 11 Ch 9 Cyclotomic + Reed-Solomon (mathematical foundations) - Paper #122 Information Substrate - Vol 14 Ch 12 substrate-CI architecture (INTERNAL register per Cal #48/#49)

— Lyra, Vol 14 Ch 8 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT